影像處理
--CH3_1

姓名:邱榮誠

學號:314512065

系級:電控碩 1

日期:2025/09/29

Q1.Please depict the histogram and graph of the assigned image "aerial_view.tif", and print out the source code?

1）**source code**:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

gray_img = cv2.imread('aerial_view.tif', cv2.IMREAD_GRAYSCALE)
cv2.imwrite('Original Image.jpg', gray_img)
img1 = gray_img.ravel()

plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.imshow(gray_img, cmap='gray')
plt.axis("off")
plt.title("Original Image")

plt.subplot(1,2,2)
plt.hist(img1, 256, [0,256], density=True)
plt.title("Histogram of Original Image")
plt.xlabel("Gray Level")
plt.ylabel("Normalized Frequency")

plt.tight_layout()
plt.show()
```

2)**depict graph**: 原始的灰階照片整體對比度偏低，許多區域（EX:建築物與地面）在灰度上差異不大，細節不夠明顯。

depict histogram:Histogram of Original Image中，橫軸0是黑色，255為白色，可以看到灰階值主要集中在30-120之間，為灰色地帶。這表示影像中的大多數像素亮度都落在「偏黑到中灰」的區域，而很少有高亮度（>200）的像素。

3)**result**:

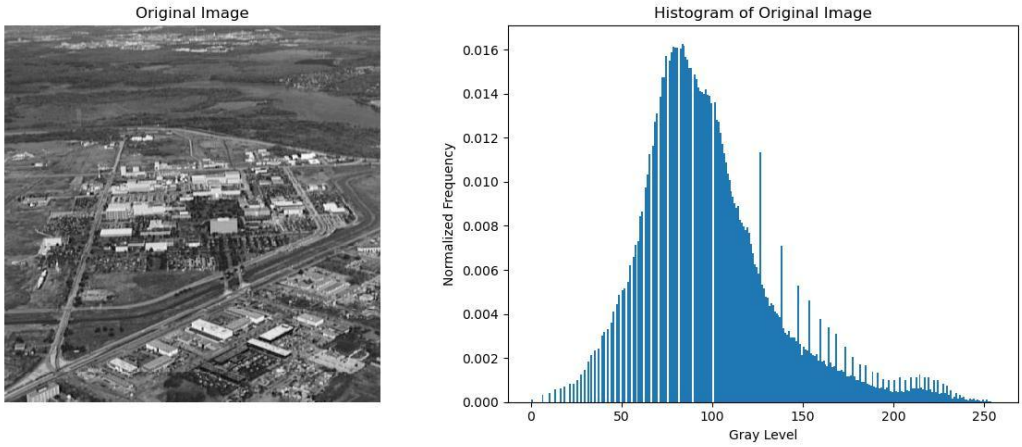

Figure1.Original Image and Histogram

Q2.Please plot the histogram and graph of the image after Histogram Equalization, and print out the source code?

1) **source code**:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread('aerial_view.tif', cv2.IMREAD_GRAYSCALE)
flat = img.ravel()
hist = np.bincount(flat, minlength=256).astype(np.float64)
pdf  = hist / flat.size
cdf  = np.cumsum(pdf)
lut  = np.round(255 * cdf).astype(np.uint8)
img_eq = lut[img]
cv2.imwrite('Equalied Image.jpg', img_eq)

plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.imshow(img_eq, cmap='gray')
plt.axis("off")
plt.title("Equalized Image")

plt.subplot(1,2,2)
plt.hist(img_eq.ravel(), bins=256, range=(0,256), density=True,
color='blue')
plt.title("Histogram of Equalized Image")
plt.xlabel("Gray Level")
```

```
plt.ylabel("Normalized Frequency")

plt.tight_layout()
plt.savefig('Equalized_with_Hist.png')
plt.show()
```

2) result:

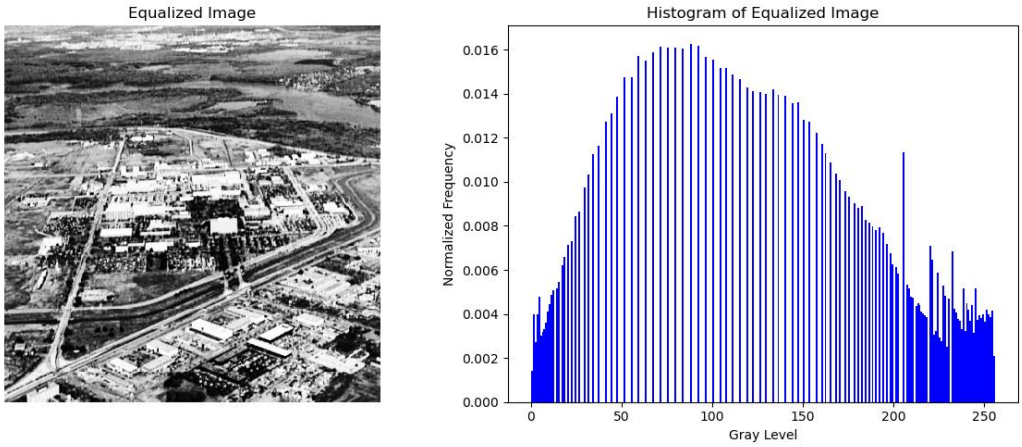

Figure2.Equalized Image and Histogram

Q3. Please plot the histogram and graph of the image after Histogram Matching (specificiation) by *pz*(*zq*)=*c*·*zq*0.4, and print out the source code? (NOTE: the parameter, *c*, needs to calculate in advance)

1) source code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('aerial_view.tif', cv2.IMREAD_GRAYSCALE)
src_flat = img.ravel()
src_hist = np.bincount(src_flat, minlength=256).astype(np.float64)
src_pdf  = src_hist / src_flat.size
src_cdf  = np.cumsum(src_pdf)
alpha = 0.4
z = np.arange(256, dtype=np.float64)
tgt_pdf = z**alpha
tgt_pdf /= tgt_pdf.sum()
tgt_cdf = np.cumsum(tgt_pdf)
lut = np.searchsorted(tgt_cdf, src_cdf, side='left').astype(np.uint8)
img_mat = lut[img]
```

```
cv2.imwrite('mat.jpg', img_mat)
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(img_mat, cmap='gray')
plt.axis('off')
plt.title('Histogram Matched Image')

plt.subplot(1,2,2)
plt.hist(img_mat.ravel(), bins=256, range=(0,256), density=True)
plt.title('Histogram of Matched Image')
plt.xlabel('Gray Level')
plt.ylabel('Normalized Frequency')

plt.tight_layout()
plt.savefig('HistogramMatched_with_Hist.png')
plt.show()
```

2)result:

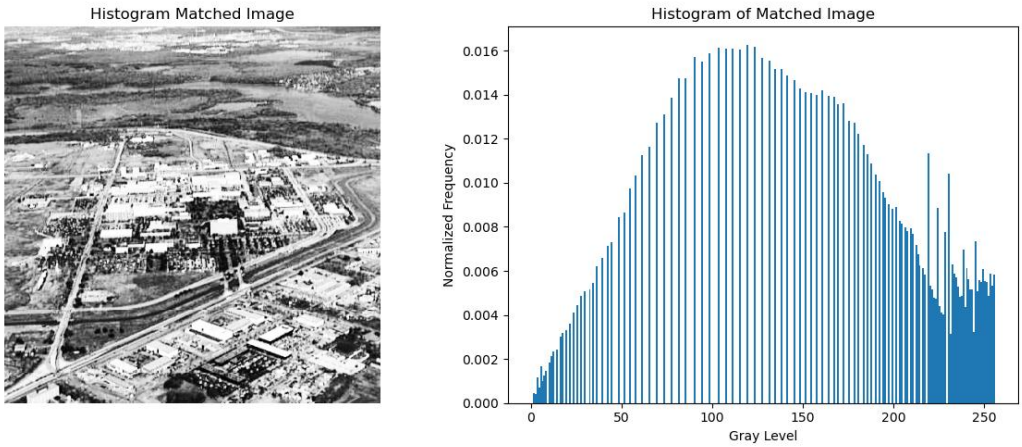

Figure3. Histogram Matched Image and Histogram

Q4. Please comment the original, the histogram-equalized and the histogram-matching images ?

Origianl經過Histogram-Equalized後，整體對比度提升。暗部變得更暗，亮部變得更亮，物體的邊界更加清晰。細節變得明顯，但同時也出現過度強化，導致某些區域看起來過亮或過暗，影像自然度下降。

Origianl經過Histogram-Matching後，對比度增加，但調整方向依照目標分布，結果比Histogram-Equalized柔和一些。畫面仍然增強了細節，整體亮度和灰階分布更有控制，效果較自然。